

$$(EE) \quad C_{t+1} = \underbrace{1}_R C_t$$

$$(BC) \quad B_{t+1} = R B_t + Y_t - C_t$$

① SS : $\begin{cases} \bar{C} = \bar{C} \\ \bar{B} = R \bar{B} + \bar{Y} - \bar{C} \end{cases} \Rightarrow \bar{C} = (R-1) \bar{B} + \bar{Y}$
 $\alpha \equiv \frac{\bar{Y}}{\bar{B}} \Rightarrow \frac{\bar{C}}{\bar{B}} = R-1 + \alpha$ $\bar{B} \neq 0$

② LL : (EE) $\Rightarrow \cancel{C}_{t+1} = \cancel{C}_t \Rightarrow \boxed{C_{t+1} = C_t}$
 (BC) $\Rightarrow \bar{B} (1 + b_{t+1}) = R \bar{B} (1 + b_t) + \bar{Y} (1 + \bar{y}_t) - \bar{C} (1 + c_t)$
 $\cancel{\bar{B}} + \bar{B} b_{t+1} = \cancel{R \bar{B}} + R \bar{B} b_t + \cancel{\bar{Y}} - \cancel{\bar{Y}} - \bar{C} \cdot c_t$
 $b_{t+1} = R b_t - \frac{\bar{C}}{\bar{B}} \cdot c_t \Rightarrow \boxed{b_{t+1} = R \cdot b_t - (R-1+\alpha) c_t}$

③ $\begin{pmatrix} b_{t+1} \\ c_{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} R & -(R-1+\alpha) \\ 0 & 1 \end{pmatrix}}_A \underbrace{\begin{pmatrix} b_t \\ c_t \end{pmatrix}}_{x_t}$ $z_{t+1} = \lambda_1 z_t$
 $x_{t+1} = \mathcal{Q} \Lambda \mathcal{Q}^{-1} x_t$
 $\underbrace{\mathcal{Q}^{-1} x_{t+1}}_{z_{t+1}} = \Lambda \underbrace{\mathcal{Q}^{-1} x_t}_{z_t}$
 $\lambda_1 = R > 1 \Rightarrow \begin{pmatrix} R & -(R-1+\alpha) \\ 0 & 1-R \end{pmatrix} v^1 = 0 \Rightarrow v^1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

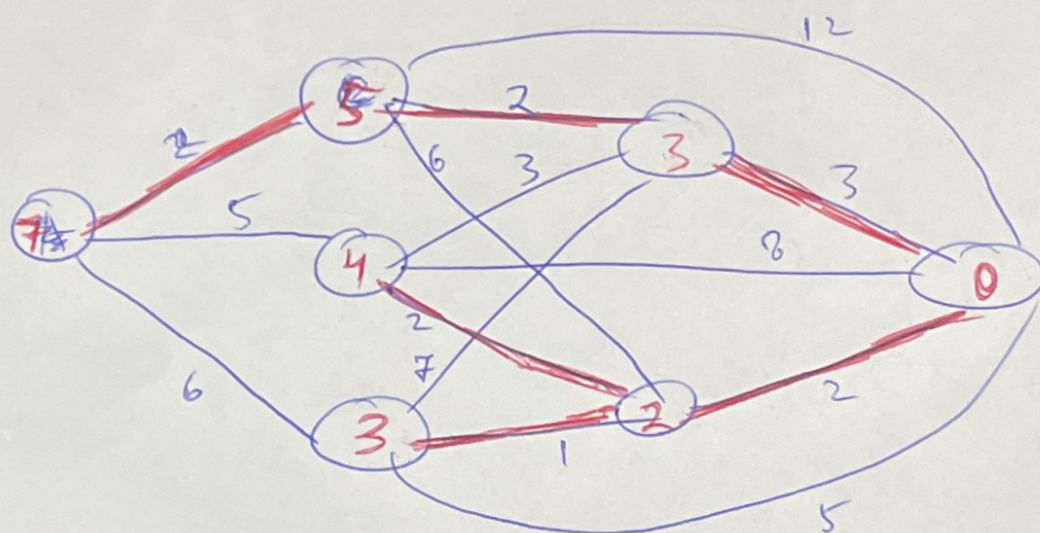
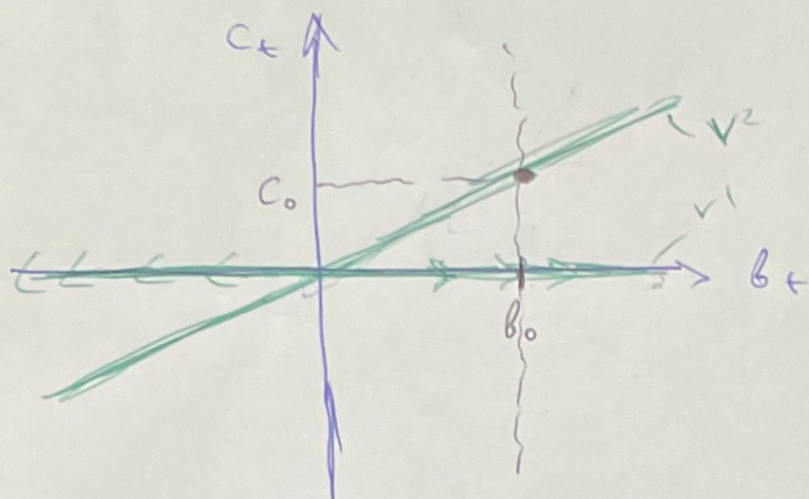
$\lambda_2 = 1 \Rightarrow \begin{pmatrix} R-1 & -(R-1+\alpha) \\ 0 & 0 \end{pmatrix} v^2 = 0 \Rightarrow v^2 = \begin{pmatrix} R-1+\alpha \\ R-1 \end{pmatrix}$

$\mathcal{Q} \equiv (v^1 \ v^2) = \begin{pmatrix} 1 & R-1+\alpha \\ 0 & R-1 \end{pmatrix} \Rightarrow \mathcal{Q}^{-1} = \frac{1}{R-1} \begin{pmatrix} R-1 & -(R-1+\alpha) \\ 0 & 1 \end{pmatrix}$

$z_{10} = 0 \Rightarrow \mathcal{Q}^{-1} x_t = 0 \Rightarrow (R-1) b_t - (R-1+\alpha) c_t = 0$

$$\Rightarrow \boxed{c_t = \frac{R-1}{R-1+\alpha} b_t} \leftarrow \text{saddle point} \quad (2)$$

$$\Rightarrow b_{t+1} = R b_t - (R-1+\alpha) c_t = R b_t - (R-1) b_t = b_t$$



$$\beta(x) = x$$

$$x = \tilde{x} \rightarrow$$

$$V(B_t) = \max_{B_{t+1}} \{ u(B_t - B_{t+1}) + \beta V(B_{t+1}) \}$$

$\uparrow$ past $\uparrow$ future

Bellman equation

~~$V(B_t)$~~

$$\underline{V(B)} = \max_{B'} \{ \log(B - B') + \beta V(B') \}$$

guess and verify $\underline{V(B)} = \alpha + \gamma \log B$; $\frac{C(B)}{B'(B)}$

$$\max_{B'} \log(B - B') + \beta 2 + \beta \gamma \log B'$$

$$FOC: \frac{-1}{B-B'} + \frac{\beta\gamma}{B'} = 0 \Rightarrow B' = \beta\gamma(B-B') \quad (3)$$

$$\Rightarrow B' = \frac{\beta\gamma}{1+\beta\gamma} B = \frac{\beta}{1-\beta+\beta} B = \boxed{\beta B} \Rightarrow \boxed{C = (1-\beta)B}$$

$$\log(B - \frac{\beta\gamma}{1+\beta\gamma} B) + \beta\alpha + \beta\gamma \log \frac{\beta\gamma}{1+\beta\gamma} B$$

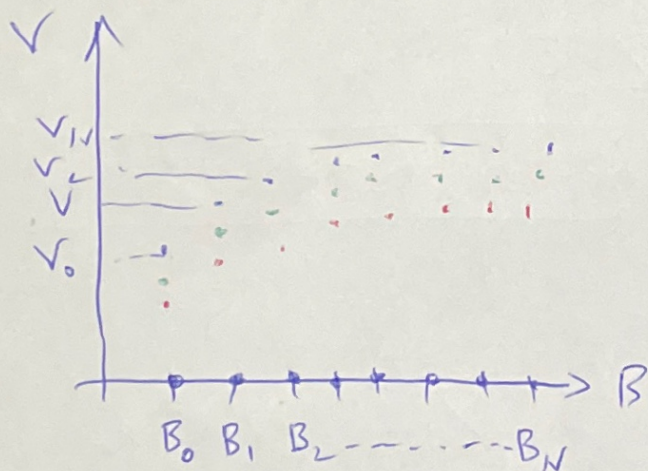
$$= \log B + \log \frac{1}{1+\beta\gamma} + \beta\alpha + \beta\gamma \log \frac{\beta\gamma}{1+\beta\gamma} + \beta\gamma \log B$$

$$= \alpha + \gamma \log B \quad \text{for } \forall B$$

$$\Rightarrow 1 + \beta\gamma = \gamma$$

$$\left\{ \log \frac{1}{1+\beta\gamma} + \beta\alpha + \beta\gamma \log \frac{\beta\gamma}{1+\beta\gamma} = \alpha \right.$$

$$\Rightarrow \gamma = \frac{1}{1-\beta}$$



$$V = \begin{pmatrix} V_0 \\ \vdots \\ V_N \end{pmatrix}$$

space of
V functions
V₁, V₂, ..., V

$$V(B) = \max_{B'} \left\{ u(\underbrace{RB + y - B'}_C) + \beta V(B') \right\}$$

$$FOC: -u'(C) + \beta V'(B') = 0$$

$$EC: V'(B) = u'(C) \cdot R \Rightarrow V'(B') = u'(C') \cdot R$$

$$u'(C) = \beta R \cdot u'(C') \Rightarrow \boxed{u'(C_e) = \beta R u'(C_{t+1})}$$

$$V(B_t) = \max_{C_t} \left\{ u(C_t) \cdot \Delta + \underbrace{e^{-\rho \Delta}}_{1-\rho \Delta} \cdot V(B_{t+\Delta}) \right\} \quad (4)$$

$$\Rightarrow \max_{C_t} \left\{ u(C_t) \Delta - \rho V(B_{t+\Delta}) \cdot \Delta + V(B_{t+\Delta}) - V(B_t) \right\} = 0$$

$$\Rightarrow \max_{C_t} \left\{ u(C_t) - \rho V(B_{t+\Delta}) + \frac{V(B_{t+\Delta}) - V(B_t)}{\Delta} \right\} = 0$$

$$\Rightarrow \rho V(B_t) = \max_{C_t} \left\{ u(C_t) + \underbrace{\frac{V(B_{t+\Delta}) - V(B_t)}{B_{t+\Delta} - B_t} \cdot \frac{B_{t+\Delta} - B_t}{\Delta}}_{\frac{dV(B)}{dB}} \right\}$$

$$\rho V(B) = \max_C \left\{ u(C) + V'(B) \cdot \dot{B} \right\} \quad (\text{HJB eq.})$$

Example: $u(C) = \log C$, $\dot{B} = 2B - C$

$$V(B) = 2 + \gamma \log B$$

$$\max_C \left\{ \log C + \frac{\gamma}{B} \cdot (2B - C) \right\}$$

$$\text{FOC: } \frac{1}{C} = \frac{\gamma}{B} \Rightarrow C = \frac{B}{\gamma}$$

$$\log \frac{B}{\gamma} + \frac{\gamma}{B} \left(2B - \frac{B}{\gamma} \right) = 1 \cdot \log \frac{B}{\gamma} + \gamma 2 - 1$$

$$= \gamma(2 + \log B), \quad \forall B$$

$$\Rightarrow \begin{cases} \gamma \gamma = 1 \\ \gamma 2 = \gamma 2 - 1 - \log \gamma \end{cases}$$

$$\Rightarrow \boxed{\gamma = \frac{1}{\gamma}} \\ \boxed{C = \gamma B}$$

$$\dot{B} = 2B - \gamma B = (2 - \gamma) B$$

$$(KJB) \quad V(B) = \max \left\{ u(c) + \underbrace{V'(B) \cdot (2B - c)}_{\dot{B}} \right\} \quad (5)$$

$$FOC: \quad u'(c) = V'(B)$$

$$EC: \quad \dot{g} V'(B) = 2 V'(B) + \underbrace{V''(B) \cdot \dot{B}}_{\text{"}}$$

$$u'(c_t) = V'(B_t) \Rightarrow u''(c) \cdot \dot{c} = V''(B) \cdot \dot{B} \quad \forall t$$

$$\Rightarrow u'(c) (g - 2) = \overbrace{u''(c) \cdot \dot{c}}^{< 0}$$

$$\Rightarrow \boxed{- \frac{u''(c)}{u'(c)} \cdot \dot{c} = 2 - g} \quad \xrightarrow{\frac{\dot{c}}{c}} \quad \left(\frac{\dot{c}}{c} = 2 - g \right)$$

Lagrangian /
Hamiltonian

Bellman eq. /
KJB eq.

optimality conditions
(EE)

Shooting
algorithm

+ non-linear
+ high dim
- only deterministic

log-linearize +
dynamic system

- no non-linearities
+ high dim
+ w/ shocks

Value
function
iteration

+ non-linear
- low dim
+ w/ shocks

policy function
iteration

$$\begin{cases} u'(c_t) = \beta \lambda u'(c_{t+1}) & C(B, y) \\ B_{t+1} = KB_t + y_t - c_t & B'(B, y) \end{cases}$$